

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.7 SURFACE AREA AND VOLUME OF SOLIDS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why does the same amount of metal sheet make containers that hold very different amounts of grain, and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? Use what you observed and calculated in Lessons 1 to 8 to write a full explanation. Answer every section in the space under the prompt. Show your working for every calculation, use $\pi = \frac{22}{7}$, and include units (cm^2 , cm^3 , m^2 , m^3 , litres). Refer to Mzee Kiplagat's posho mill in Eldoret, Wanjiru the jua kali metalsmith, and your own class evidence. Remember: 1 litre = 1,000 cm^3 and 1 m^3 = 1,000 litres.

Part 1: The puzzle and the box

Mzee Kiplagat's posho mill in Eldoret has two closed containers made by Wanjiru: a cylindrical drum and a square-based box. Wanjiru says she used exactly 30,800 cm^2 of iron sheet for each. (a) Explain how you can use a net to find the total surface area of a prism. (b) The box measures 50 cm x 50 cm x 129 cm. Calculate its total surface area and say whether it agrees with Wanjiru's figure. (c) Explain why knowing the surface area alone does not tell Mzee Kiplagat how much maize a container holds.

Write your answer here:

Part 2: Testing the drum

The drum has radius 35 cm and height 105 cm. (a) Decompose the closed drum into its parts and calculate its total surface area. Does the method reproduce Wanjiru's figure? (b) Calculate the volume of the drum and the volume of the box in cm^3 , then convert each to litres. (c) Which container holds more maize, and by how many litres? Does this match what Mzee Kiplagat saw?

Write your answer here:

Part 3: Explaining why shape matters

Use your evidence from Parts 1 and 2 to explain the main puzzle. (a) Complete this idea in your own words: 'Equal surface area does not mean equal volume because...'. (b) Using the cross-section of each container, explain why the drum's shape encloses more space than the box's shape for the same amount of sheet. (c) Of the solids you studied, which shape would hold the most for a given surface area? Explain why a sphere is not always the best choice for a real grain store.

Write your answer here:

Part 4: Surface area question or volume question?

The Rift Valley cooperative wants a water tank made of a cylinder (radius 2.1 m, height 4 m, flat base) with a hemisphere on top, and a grain silo made of a cylinder (radius 2.1 m, height 6 m) with a cone roof. Mzee Kiplagat also has a pyramid roof (6 m x 6 m base, vertical height 4 m) on a small store. (a) For each of these jobs, say whether you need surface area or volume: (i) painting the tank, (ii) finding how much water the tank holds, (iii) finding how much iron sheet covers the pyramid roof (no floor), (iv) finding the space enclosed by the pyramid roof. (b) Calculate the roof sheet needed in (iii) and the volume in (iv). (c) When you find the surface area of a composite solid such as the tank, what must you do about the circle where two parts join? Why?

Write your answer here:

Part 5: Using mathematics to design

Answer the KICD inquiry questions using your own project or the class evidence. (a) How do we determine the surface area and volume of solids? Give at least three different solids and the formula or method you used for each. (b) How are surface area and volume applied in real life in Kenya? Describe one design for a named Kenyan community need (for example a grain silo, water tank or sufuria), say which calculation you would do for sheet or paint and which for capacity, and name one career that uses this mathematics. (c) Finally, write a short final explanation (3 to 5 sentences) answering the driving question.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Calculating surface area and volume correctly	Uses the correct method for each solid (prism, cylinder, pyramid, cone, sphere, composite), uses slant height and vertical height correctly, gets all values right (30,800 cm ² ; 404,250 cm ³ ; 322,500 cm ³) with correct units and working.	Uses mostly correct methods and gets most values right, with one or two small slips in arithmetic, units or choice of height.	Confuses formulae, mixes slant and vertical height, or omits working and units, so the values are mostly incorrect.
Distinguishing surface area from volume	States clearly that surface area is the material covering a solid and volume is the space inside, and correctly classifies each task as a surface area or volume question with reasons.	Distinguishes the two ideas correctly but gives limited reasons or misclassifies one task.	Treats surface area and volume as the same thing or cannot say which one a task needs.
Explaining the phenomenon with evidence	Explains why the drum holds 81.75 litres more than the box using the shared 30,800 cm ² , both volumes and the cross-section areas (3,850 cm ² against 2,500 cm ²), noting that the drum's larger base outweighs the box's extra height, and links to the sphere as the most efficient shape.	Explains that shape decides capacity and quotes the main numbers, but the reasoning about cross-sections is incomplete.	States that the drum holds more without evidence, or suggests that equal surface area means equal volume.

Handling composite solids and real-life application	Leaves out hidden joined faces in surface area, adds the volumes of the parts, and describes a realistic Kenyan design, explaining which calculation gives sheet and which gives capacity, plus a relevant career.	Handles composite solids mostly correctly and gives a reasonable Kenyan application with some detail missing.	Counts hidden faces or gives no clear real-life application.
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ATTRIBUTION AND LICENSING

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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